Aprendizaje Automático II

Dr. Federico Albanese
Franco Castellacci


Ceci n'est pas une pipe.

¿Qué significan las palabras?





Significante
[R-O-S-A]

Significado
[Flor roja]

¿Cómo pasar del significante al significado?



Bag of words

¿Cúal es el método más simple?

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Contar términos que aparecen en los textos

Bag of words

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Ejemplo:

John likes to watch movies. Mary likes movies too.

John also likes to watch football games.

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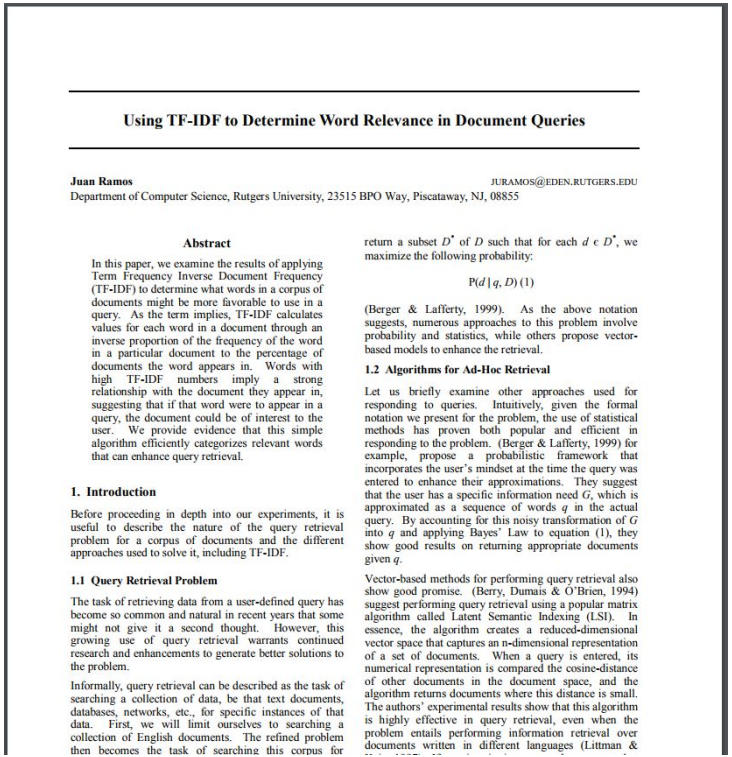
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¿Cuando sirve esto? -> Ej. Clasificar textos, Spam (Hay términos que de aparecer definen la clase)

TF-IDF: Se puede hacer un poco mejor



Using TF-IDF to Determine Word Relevance in Document Queries

Juan Ramos JURAMOS@EDEN.RUTGERS.EDU
Department of Computer Science, Rutgers University, 23515 BPO Way, Piscataway, NJ, 08855

Abstract

In this paper, we examine the results of applying Term Frequency Inverse Document Frequency (TF-IDF) to determine what words in a corpus of documents might be more favorable to use in a query. As the term implies, TF-IDF calculates values for each word in a document through an inverse proportion of the frequency of the word in a particular document to the percentage of documents the word appears in. Words with high TF-IDF numbers imply a strong relationship with the document they appear in, suggesting that if that word were to appear in a query, the document could be of interest to the user. We provide evidence that this simple algorithm efficiently categorizes relevant words that can enhance query retrieval.

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1.1 Query Retrieval Problem

The task of retrieving data from a user-defined query has become so common and natural in recent years that some might not give it a second thought. However, this growing use of query retrieval warrants continued research and enhancements to generate better solutions to the problem.

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1.2 Algorithms for Ad-Hoc Retrieval

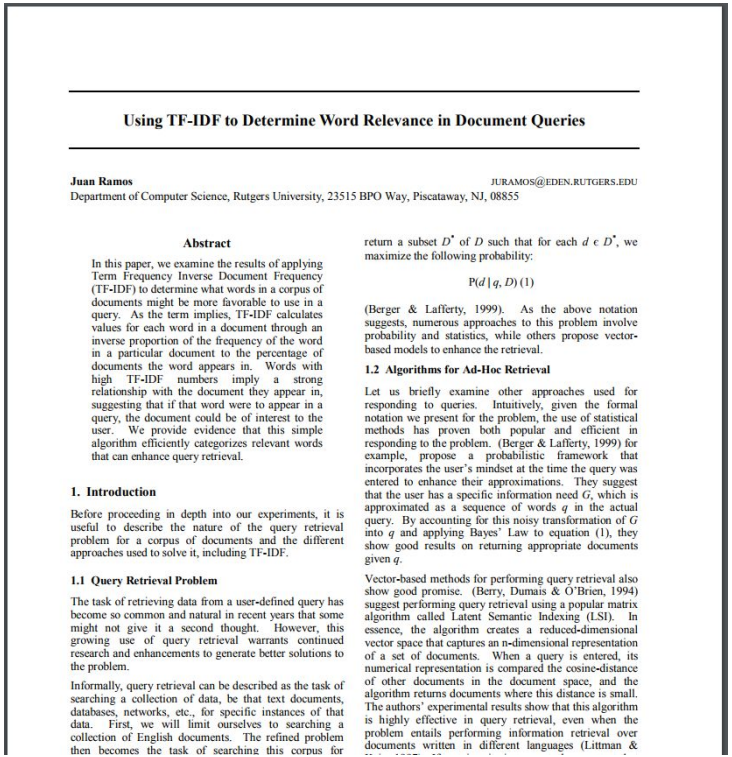
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Vector-based methods for performing query retrieval also show good promise. (Berry, Dumais & O'Brien, 1994) suggest performing query retrieval using a popular matrix algorithm called Latent Semantic Indexing (LSI). In essence, the algorithm creates a reduced-dimensional vector space that captures an n-dimensional representation of a set of documents. When a query is entered, its numerical representation is compared the cosine-distance of other documents in the document space, and the algorithm returns documents where this distance is small. The authors' experimental results show that this algorithm is highly effective in query retrieval, even when the problem entails performing information retrieval over documents written in different languages (Littman &

Cada Texto se representa vectorialmente

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con:

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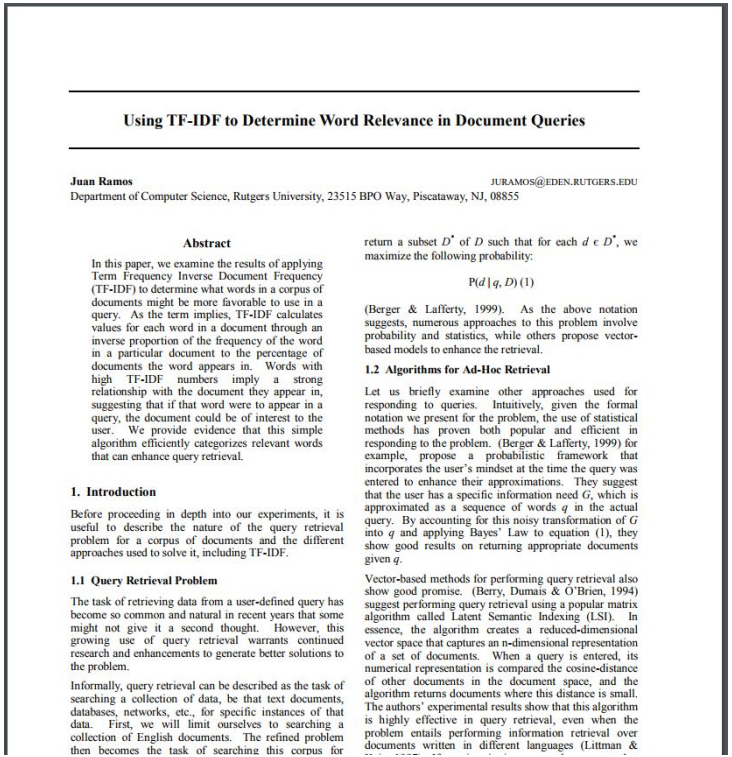
$$IDF = 1 + log(\frac{1+N}{1+n_t})$$

donde N es el # total de documentos y nt el # de documentos donde aparece el término t.

La dimensión de V está dada por el # de términos.

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donde N es el # total de documentos y nt el # de documentos donde aparece el término t.

La dimensión de V está dada por el # de términos.

De esta forma obtenemos una matriz M con la información de todos los documentos en donde cada fila es un texto y cada columna un término.

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Reducción de la dimensión: ML no supervisado

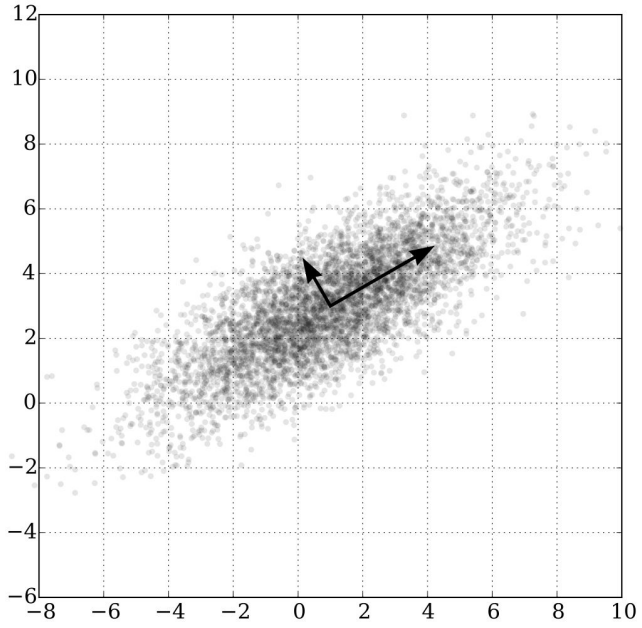
Dado un corpus de textos representado como una matriz con valores numéricos, busco reducir la dimensionalidad: tener menos columnas.

$\tilde{A} = A * \tilde{U}$ donde $A:(N*M)$; $\tilde{A}(N*D)$; $\tilde{U}:(M*D)$ con $D<M$

$\tilde{U}$ es mi matriz de cambio de base

¿Cómo sería la nueva matriz? ¿Qué serían mis nuevas columnas?

PCA: Principal component analysis



Dado un corpus de textos representado como una matriz con valores numéricos, busca una rotación en los ejes de manera de maximizar la varianza (resolver autovalores y autovectores de la matriz).

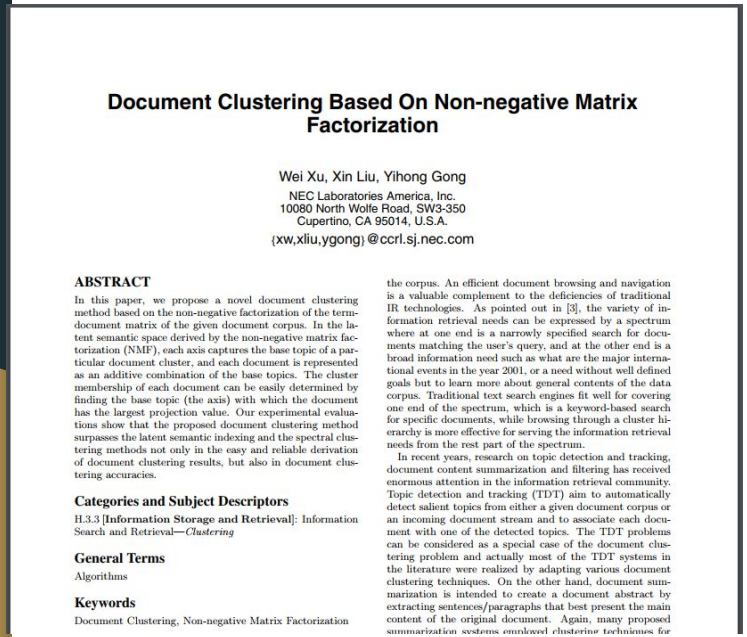
Asume que los datos pueden ser representados en una dimensión de espacio menor.

La dirección en donde se maximiza la varianza es la del autovector s cuyo autovalor sea mayor.

Proyectar: $z(i) = \tilde{U}^t x(i)$

Reconstruir: $x(i) = \tilde{U}\, z(i)$

Factorización no negativa de matrices (NMF)



Document Clustering Based On Non-negative Matrix Factorization

Wei Xu, Xin Liu, Yihong Gong
NEC Laboratories America, Inc.
10080 North Wolfe Road, SW3-350
Cupertino, CA 95014, U.S.A.
{xw,xliu,ygong}@ccrl.sj.nec.com

ABSTRACT

In this paper, we propose a novel document clustering method based on the non-negative factorization of the term-document matrix of the given document corpus. In the latent semantic space derived by the non-negative matrix factorization (NMF), each axis captures the base topic of a particular document cluster, and each document is represented as an additive combination of the base topics. The cluster membership of each document can be easily determined by finding the base topic (the axis) with which the document has the largest projection value. Our experimental evaluations show that the proposed document clustering method surpasses the latent semantic indexing and the spectral clustering methods not only in the easy and reliable derivation of document clustering results, but also in document clustering accuracies.

Categories and Subject Descriptors

H.3.3 [**Information Storage and Retrieval**]: Information Search and Retrieval—*Clustering*

General Terms

Algorithms

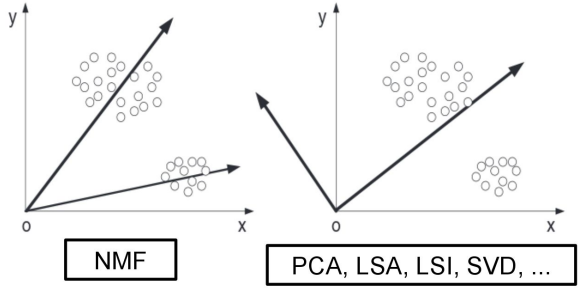
Keywords

Document Clustering, Non-negative Matrix Factorization

the corpus. An efficient document browsing and navigation is a valuable complement to the deficiencies of traditional IR technologies. As pointed out in [3], the variety of information retrieval needs can be expressed by a spectrum where at one end is a narrowly specified search for documents matching the user's query, and at the other end is a broad information need such as what are the major international events in the year 2001, or a need without well defined goals but to learn more about general contents of the data corpus. Traditional text search engines fit well for covering one end of the spectrum, which is a keyword-based search for specific documents, while browsing through a cluster hierarchy is more effective for serving the information retrieval needs from the rest part of the spectrum.

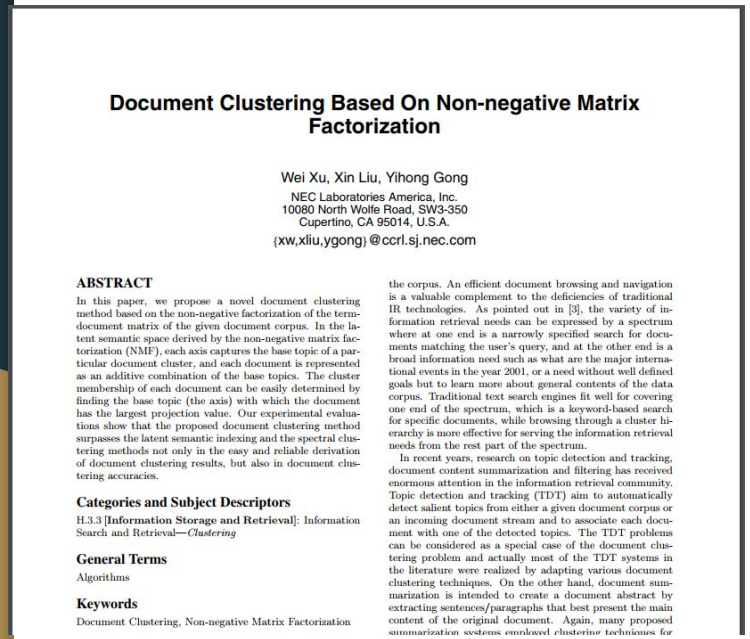
In recent years, research on topic detection and tracking, document content summarization and filtering has received enormous attention in the information retrieval community. Topic detection and tracking (TDT) aim to automatically detect salient topics from either a given document corpus or an incoming document stream and to associate each document with one of the detected topics. The TDT problems can be considered as a special case of the document clustering problem and actually most of the TDT systems in the literature were realized by adapting various document clustering techniques. On the other hand, document summarization is intended to create a document abstract by extracting sentences/paragraphs that best present the main content of the original document. Again, many proposed

NMF es una técnica de reducción dimensional, que descompone una matriz como multiplicación de dos matrices M ≈ H*W donde H es una matriz de documentos por tópico y W una matriz de tópicos por término.



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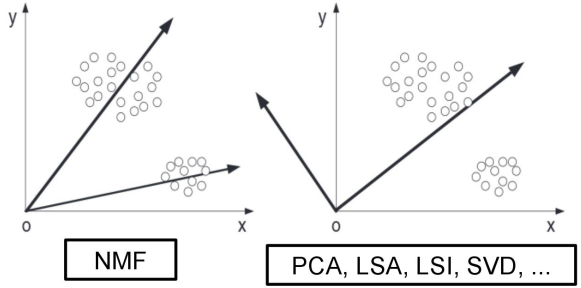
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Ventajas:

- Vectores con componentes positivas (fácil interpretación)
- Componentes no necesariamente ortogonales

Desventajas:

- Hay que elegir el # de tópicos (al igual que LSA)

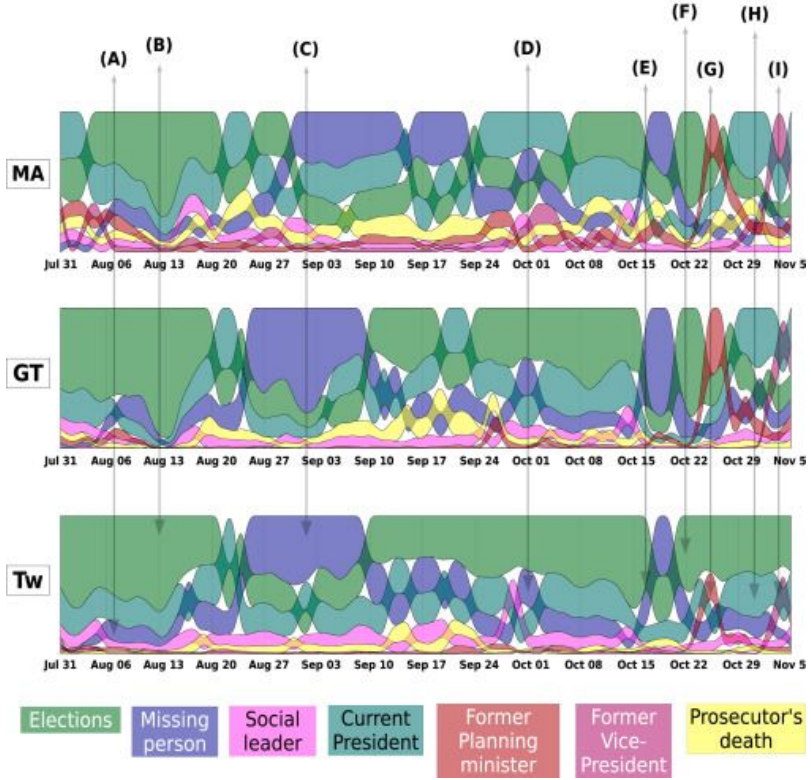
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Aplicación: ¿De que se habla en argentina?



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Word2Vec

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Embeddings de palabras

- **Skip-gram:**
  Tarea: Predecir el contexto a partir una palabra

- **CBOW**
  Tarea: Predecir una palabra a partir de las anteriores y siguientes.

Resultado: Representación vectorial de las palabras.

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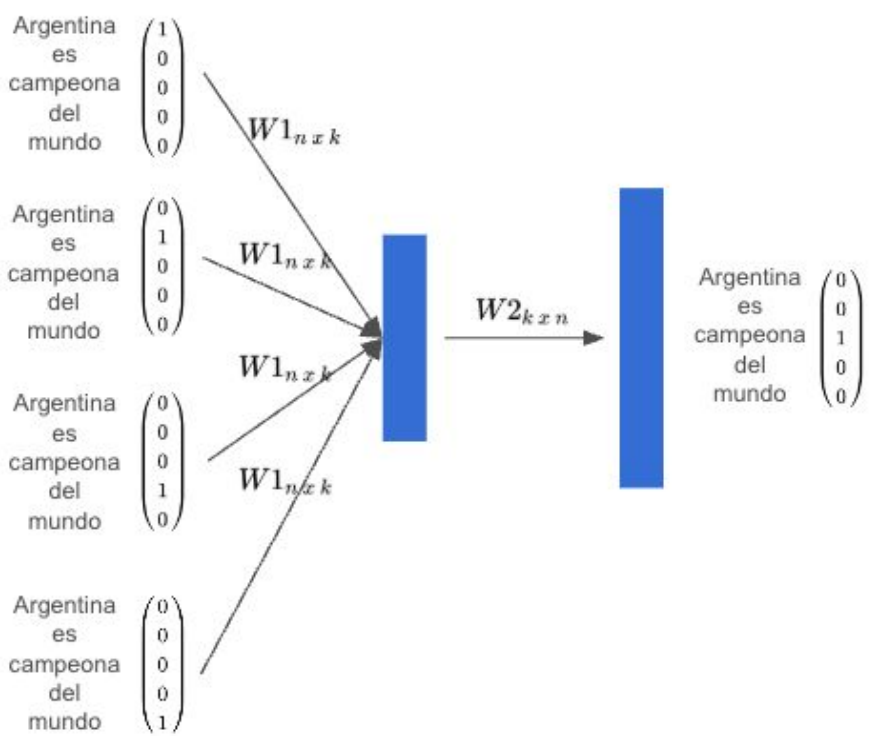
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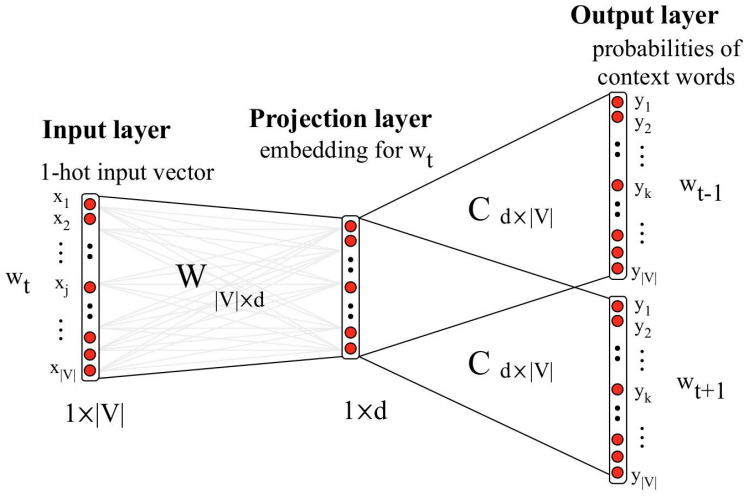


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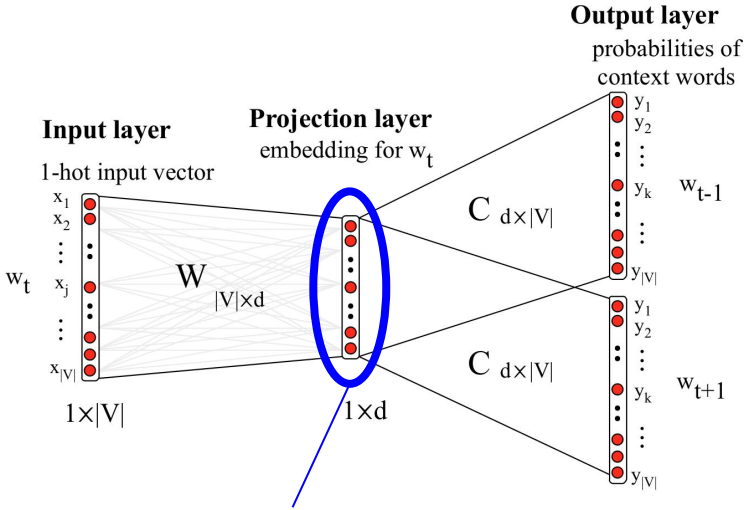
WIKIPEDIA
The Free Encyclopedia



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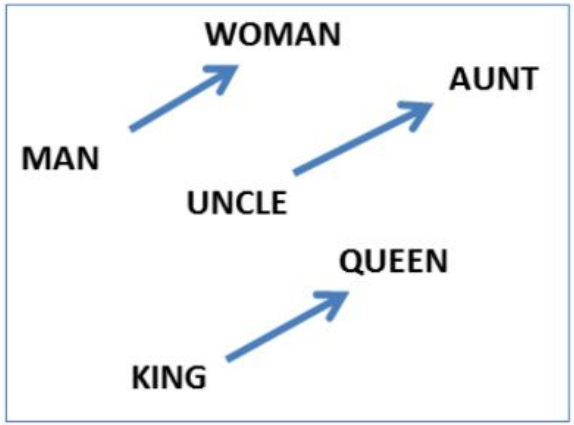
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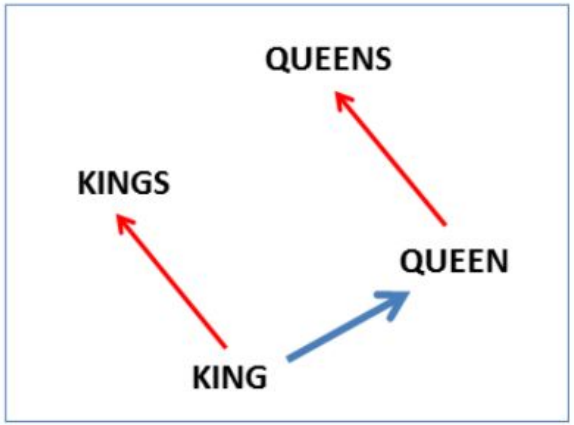
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Me quedo con esto:
Representación vectorial de dimensión d de un término

Word2Vec





$$King - Queen \approx Woman - Man$$

$$King \approx Woman - Man + Queen$$

Linguistic Regularities in Continuous Space Word Representations Mikolov et al. (2013)

T-sne: ¿Y cómo lo visualizo?

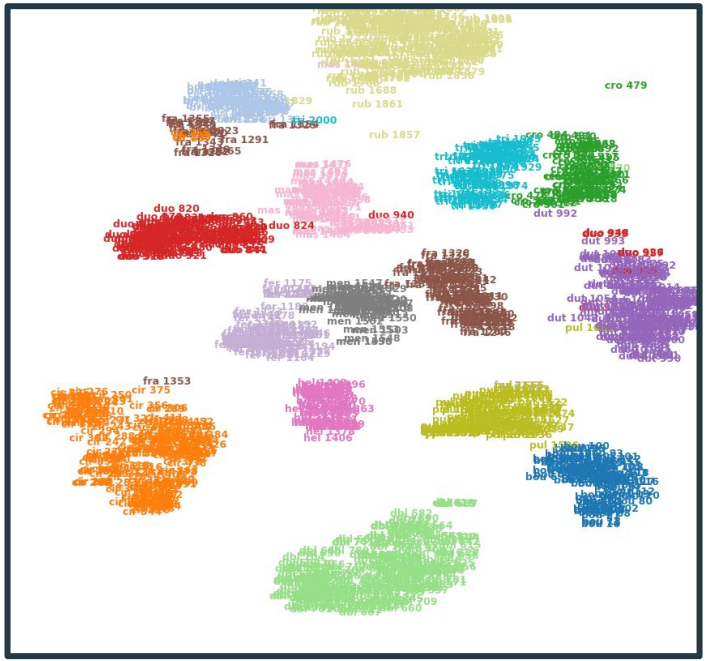
T-distributed Stochastic Neighbor Embedding

¿Qué es?

Técnica de reducción dimensional.

¿Cómo funciona?

Busca preservar la estructura local de los datos



T-sne: ¿Y cómo lo visualizo?

Pasos:

1. Convertir las similitudes entre puntos en probabilidad de elegir un punto como vecino de otro.

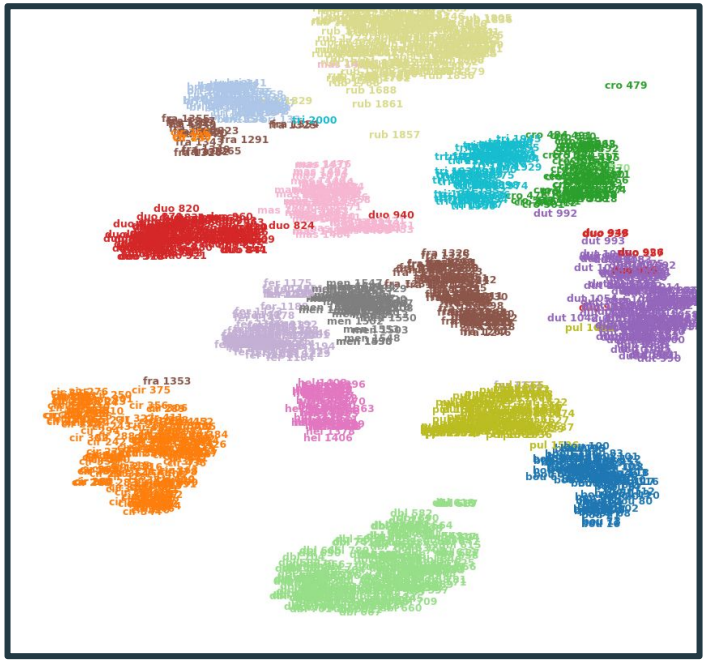
$$p_{j|i} = \frac{\exp(-\|\mathbf{x}_i - \mathbf{x}_j\|^2 / 2\sigma_i^2)}{\sum_{k \neq i} \exp(-\|\mathbf{x}_i - \mathbf{x}_k\|^2 / 2\sigma_i^2)},$$

2. Proyectar los puntos en la dimensión baja.
3. Convertir similitudes en probabilidades

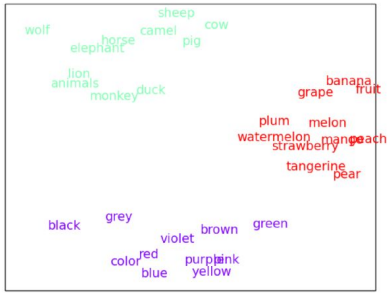
$$q_{ij} = \frac{(1 + \|\mathbf{y}_i - \mathbf{y}_j\|^2)^{-1}}{\sum_{k \neq l} (1 + \|\mathbf{y}_k - \mathbf{y}_l\|^2)^{-1}}$$

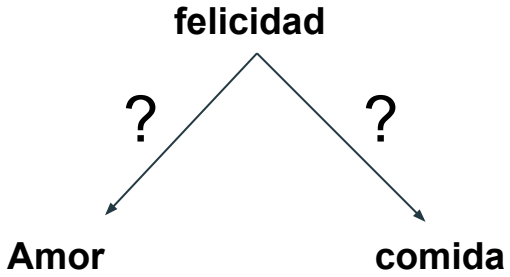
4. Minimizar (mediante descenso de gradiente)

$$Costo = \sum_i KL(P_i || Q_i) = \sum_i \sum_j p_{j|i} log(\frac{p_{j|i}}{q_{j|i}})$$



Ejemplo 1: ¿Qué son los Sueños?





Ejemplo 1: ¿Qué son los Sueños?

Más de 20.000 reportes de sueños





Schneider, A., & Domhoff, G. W. (2016). The Quantitative Study of Dreams.

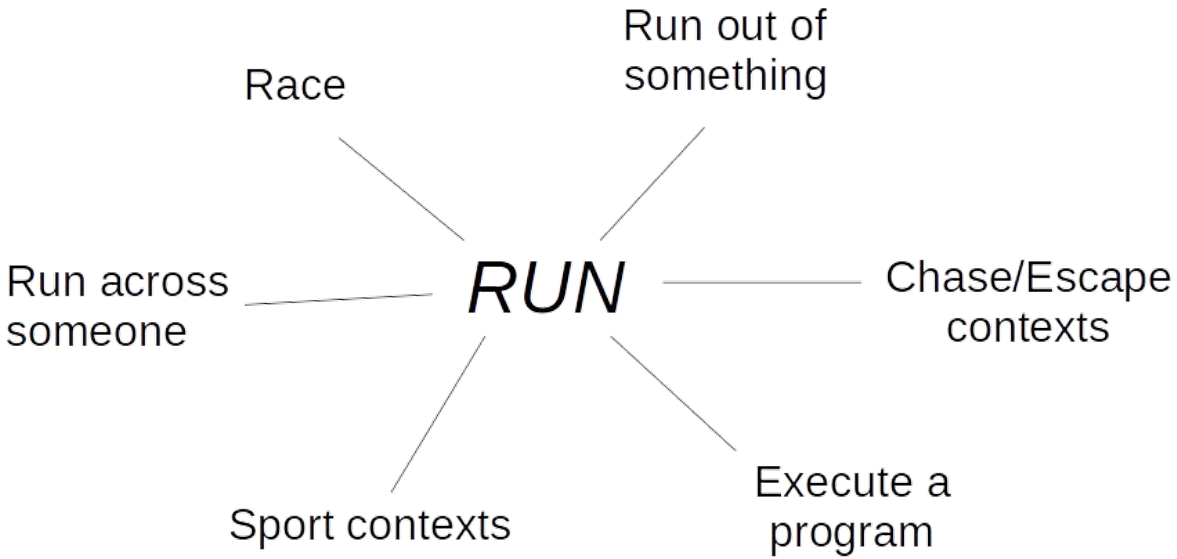
Ejemplo 1: ¿Qué son los Sueños?



Antti Revonsuo (2000) The reinterpretation of dreams: An evolutionary hypothesis of function of dreaming

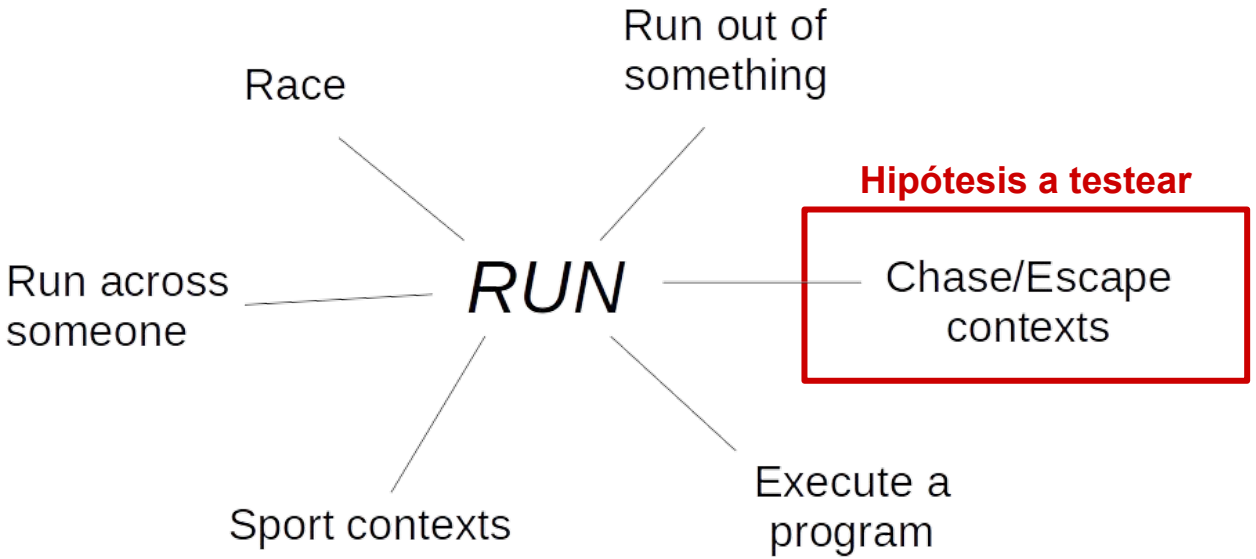
Ejemplo 1: ¿Qué son los Sueños?

¿De qué corremos en los sueños?



Ejemplo 1: ¿Qué son los Sueños?

¿De qué corremos en los sueños?



Ejemplo 1: ¿Qué son los Sueños?

Palabras más cercanas a “run”

Rank	LSA Dreams	Word2vec Dreams	LSA UkWaC	Word2vec UkWaC
1st	running	chase	running	running
2nd	escape	running	dash	runs
3rd	catch	scream	jumping	marathon
4th	chase	chasing	jump	bash
5th	chasing	escape	yell	start
6th	follow	runs	workouts	rlogin
7th	ran	chases	kick	runners
8th	sight	grab	jogging	starts
9th	coming	screaming	workout	jump
10th	runs	nazi	stretch	loaded
11th	dangerous	hide	tiring	weekend
12th	guards	chased	fun	vms
13th	robbers	yells	fast	startx
14th	hide	safety	repetitions	marathons
15th	toward	wolf	throw	mkdir

Ejemplo 2: Similaridad semántica

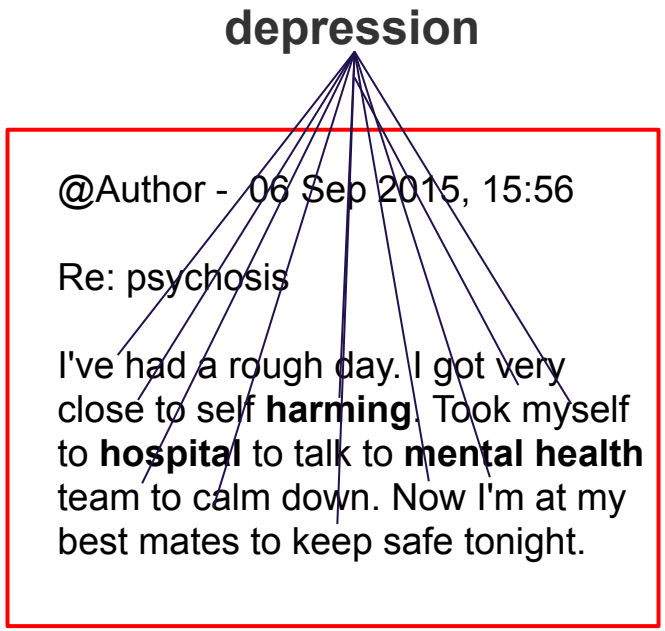
Ejemplo 2: Similaridad semántica

@Author - 06 Sep 2015, 15:56

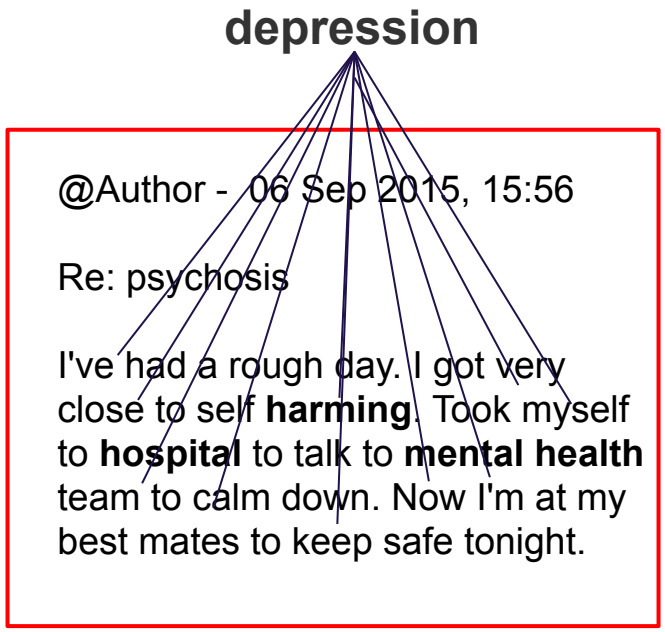
Re: psychosis

I've had a rough day. I got very close to self **harming**. Took myself to **hospital** to talk to **mental health** team to calm down. Now I'm at my best mates to keep safe tonight.

Ejemplo 2: similaridad semántica

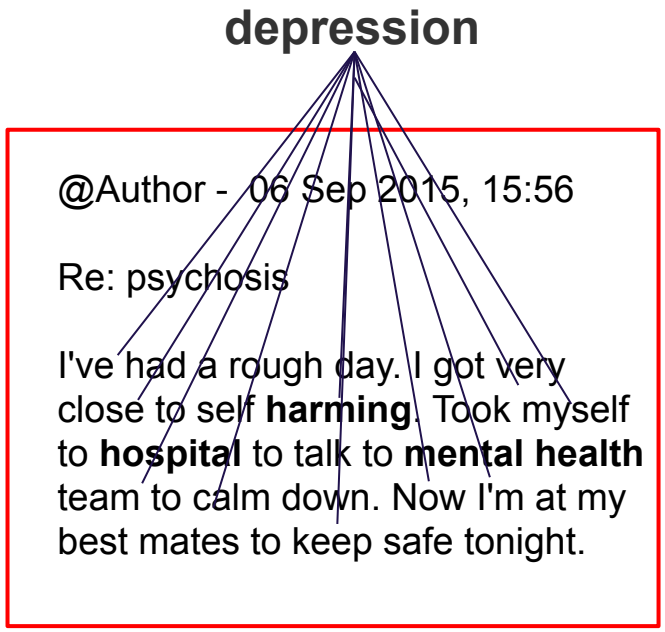


Ejemplo 2: Similaridad semántica



- Cercanía promedio = 0.05
- Fracción de palabras cercanas= 4/35

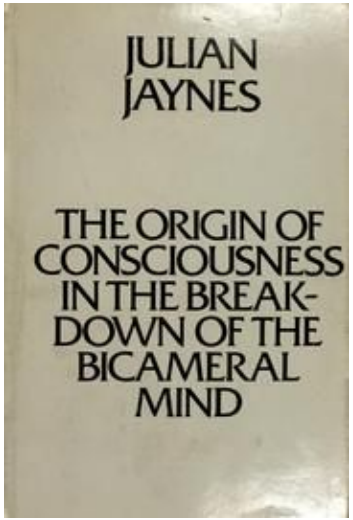
Ejemplo 2: Similaridad semántica



- Cercanía promedio = 0.05
- Fracción de palabras cercanas= 4/35

Se utilizan varias palabras:
depression - fear - anxiety - mental_health - hopelessness - suicide - antidepressant

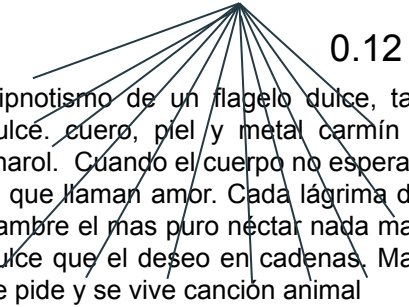
Ejemplo 2: Similaridad semántica



Hipótesis: cambios en la introspección a través del tiempo

Ejemplo 2: La historia de la introspección

Introspección

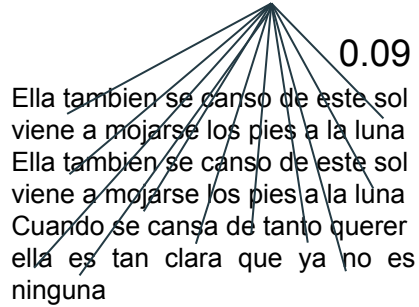


0.12

Hipnotismo de un flagelo dulce, tan dulce. cuero, piel y metal carmín y charol. Cuando el cuerpo no espera lo que llaman amor. Cada lágrima de hambre el mas puro néctar nada mas dulce que el deseo en cadenas. Mas se pide y se vive canción animal

Libro 1

Introspección



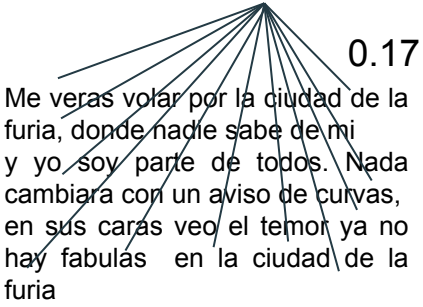
0.09

Ella tambien se canso de este sol viene a mojarse los pies a la luna Ella también se canso de este sol viene a mojarse los pies a la luna Cuando se cansa de tanto querer ella es tan clara que ya no es ninguna

Libro 2

...

Introspección

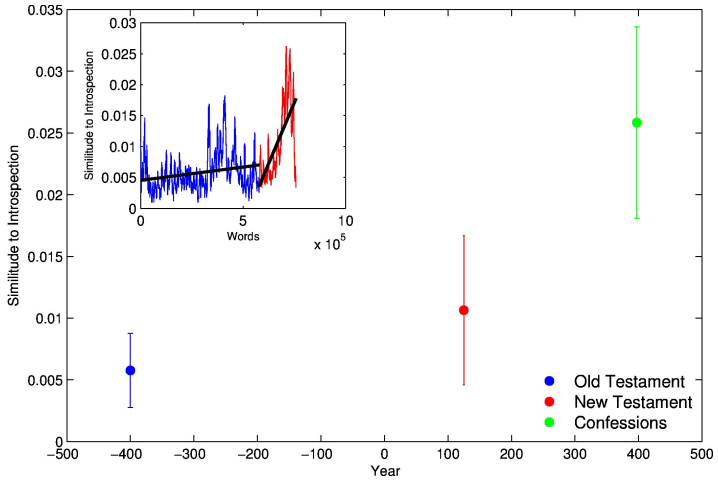


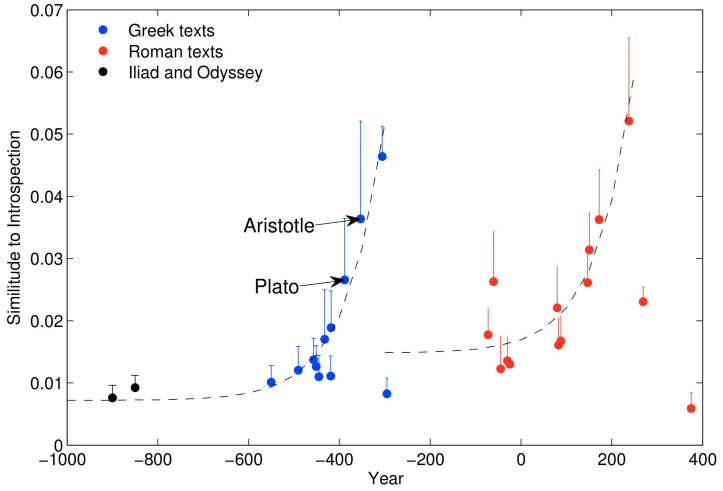
0.17

Me veras volar por la ciudad de la furia, donde nadie sabe de mi y yo soy parte de todos. Nada cambiara con un aviso de curvas, en sus caras veo el temor ya no hay fabulas en la ciudad de la furia

Libro N

Ejemplo 2: La historia de la introspección



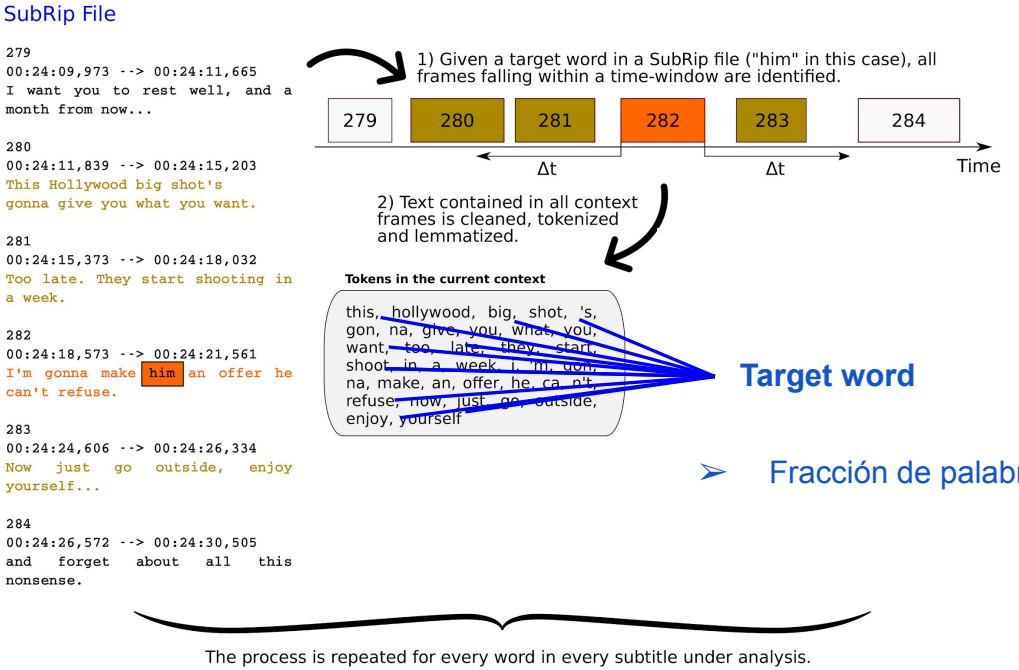


"A quantitative philology of introspection." Diuk, Carlos G., D. Fernandez Slezak, I. Raskovsky, M. Sigman, and G. A. Cecchi. (2012).

Ejemplo 2: Similaridad semántica en películas



Ejemplo 2: Similaridad semántica en películas

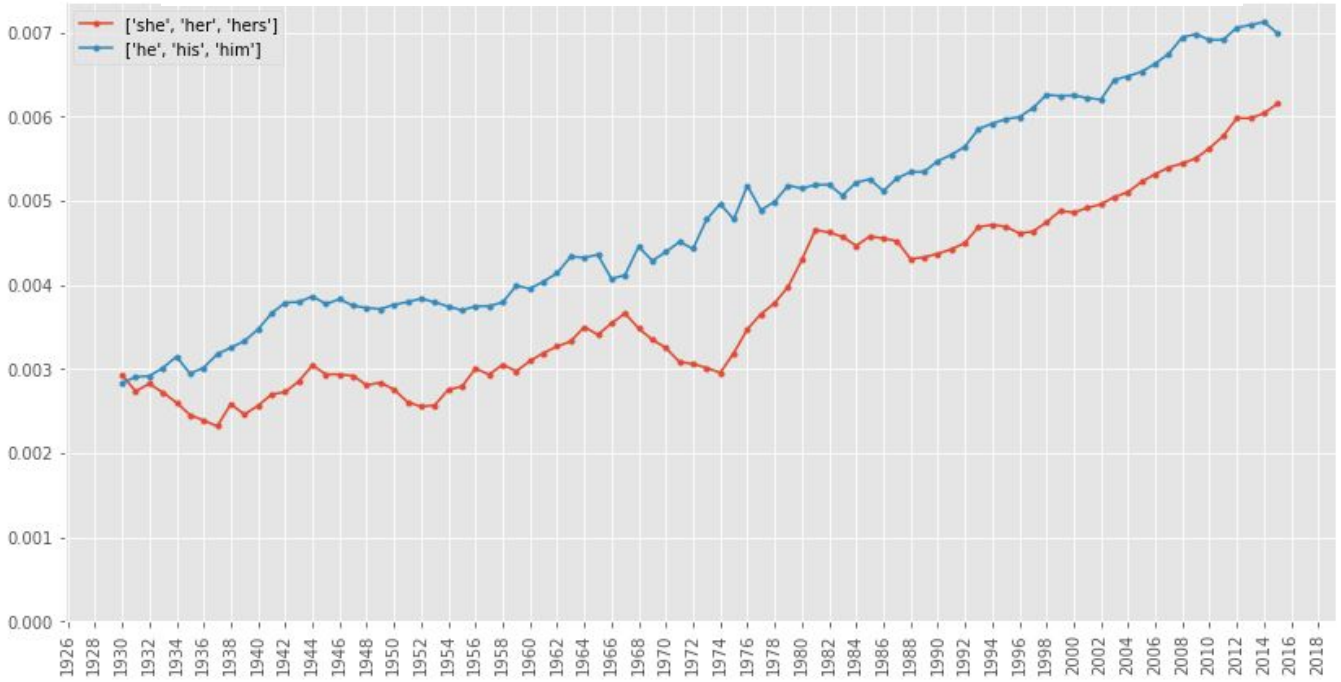


➢ Fracción de palabras cercanas = 0.20

Gálvez H. R., Tiffemberg V. and Altszyler E. (2018) Half a century of stereotyping associations between gender and intellectual ability in films

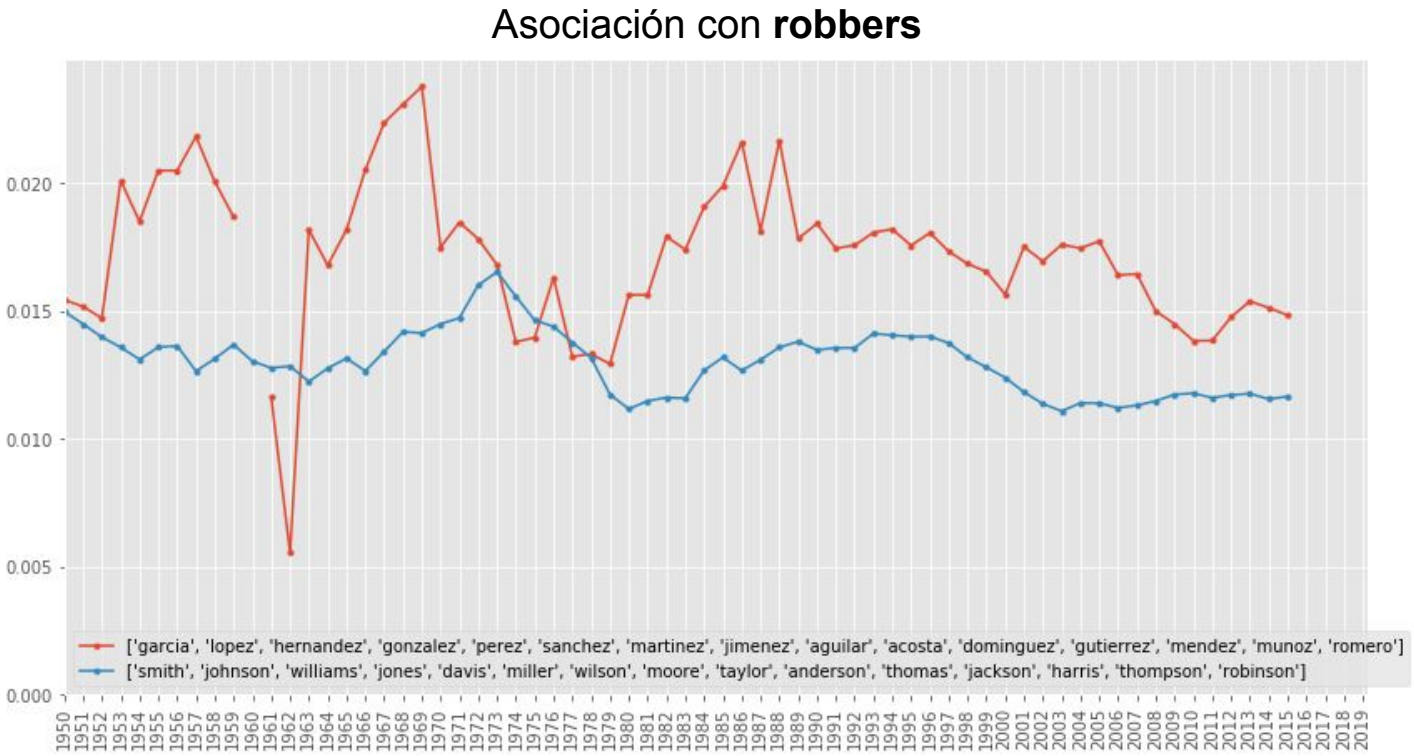
Ejemplo 2: Similaridad semántica en películas

Asociación con **technology**



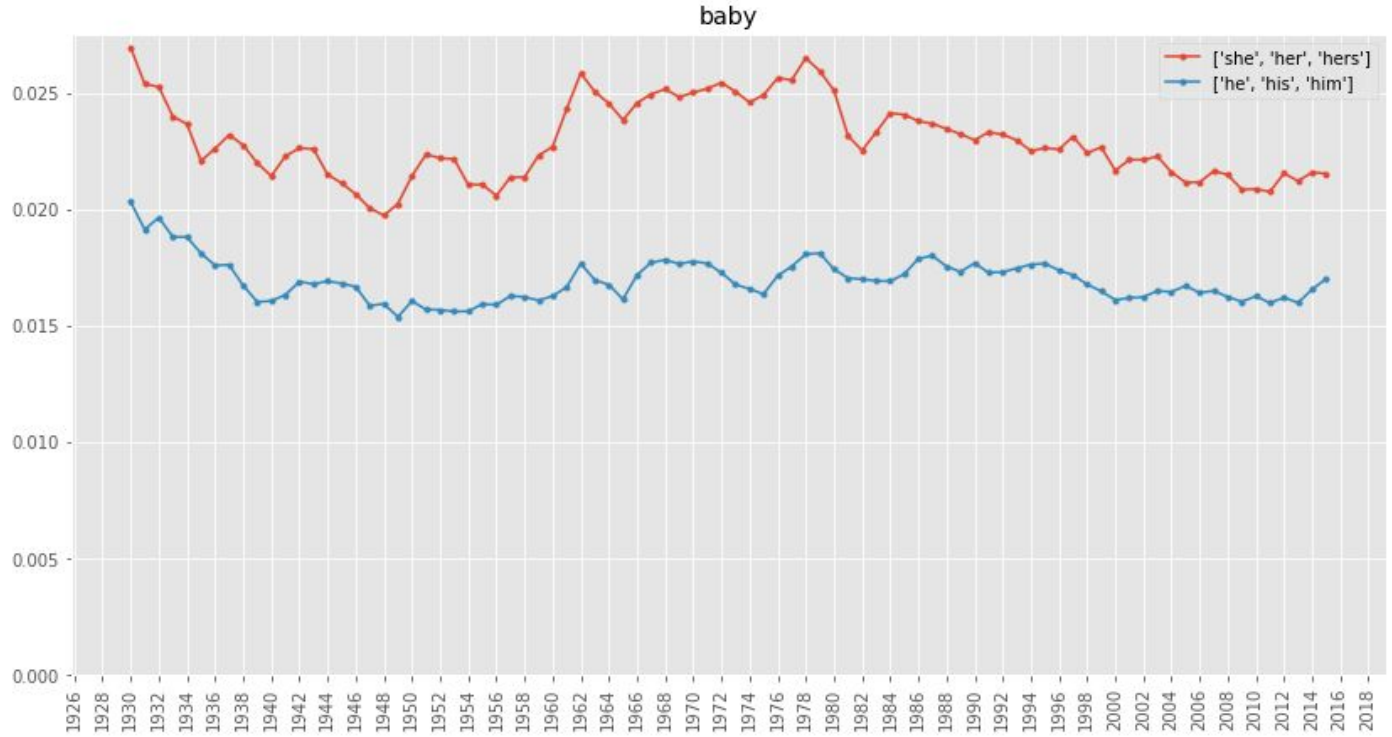
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Ejemplo 2: Similaridad semántica en películas



Gálvez H. R., Tiffemberg V. and Altszyler E. (2018) Half a century of stereotyping associations between gender and intellectual ability in films

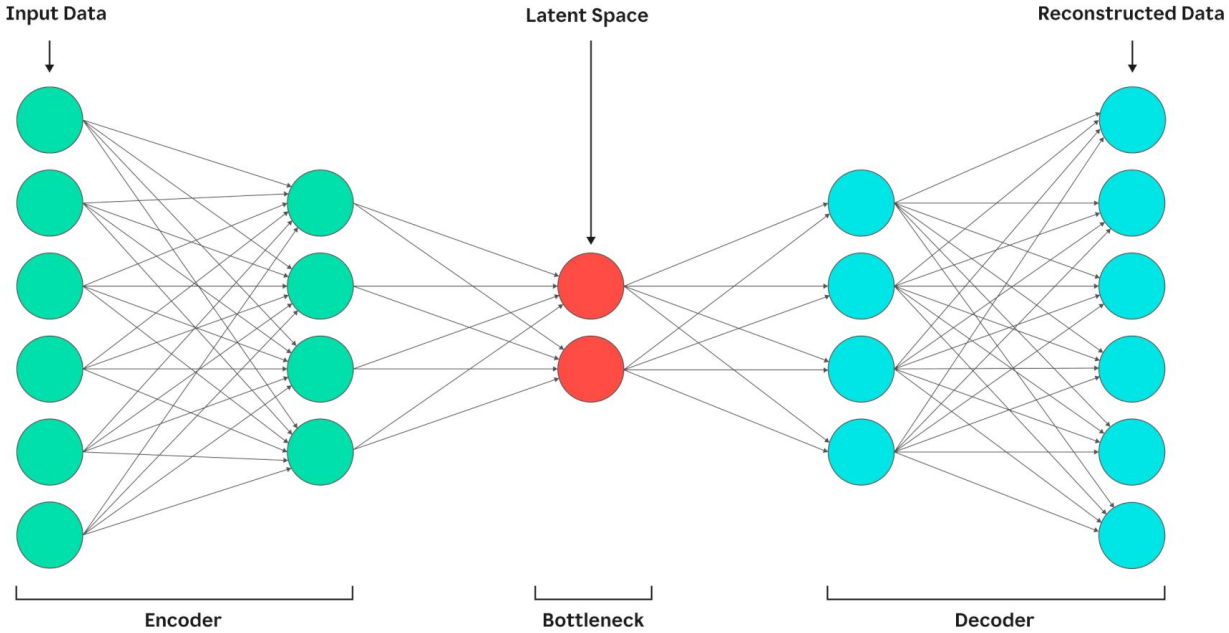
Ejemplo 2: Similaridad semántica en películas



Gálvez H. R., Tiffemberg V. and Altszyler E. (2018) Half a century of stereotyping associations between gender and intellectual ability in films

Autoencoders

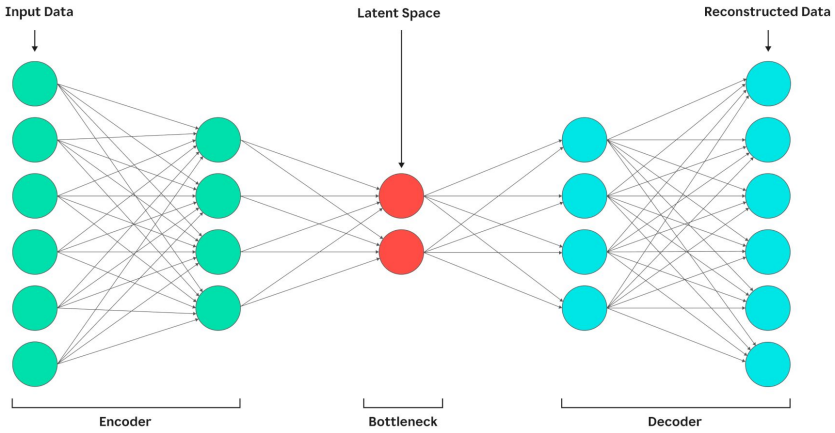
Autoencoders



Autoencoders

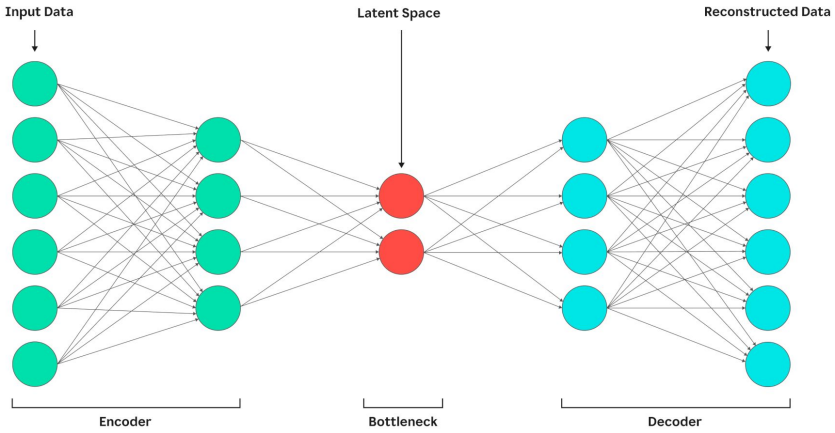
Estructura:
- Encoder
- Decoder

Objetivo:
- Reducción dimensional
- Information retrieval
- Útil para modelos generativos



Autoencoders

El latent space, de dimensión h, suele ser considerablemente menor a la dimensión del Input.

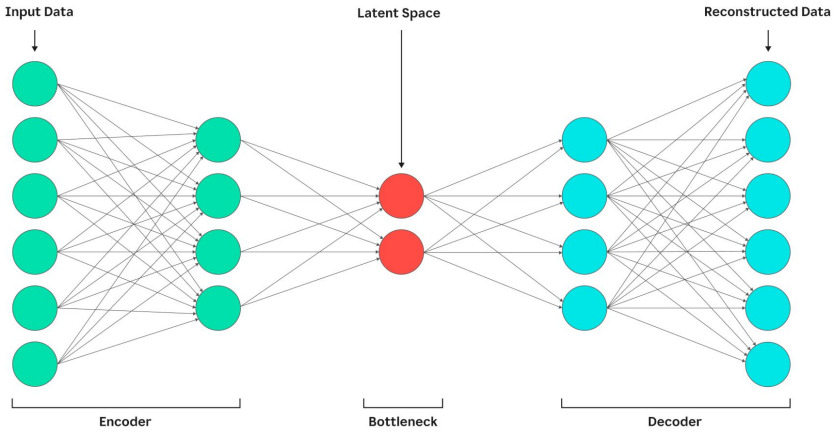


Autoencoders

Se minimiza la función de pérdida:

$$L(x, g(f(x)))$$

Donde *L* es la función de costo y se penaliza por diferenciar el output de x.



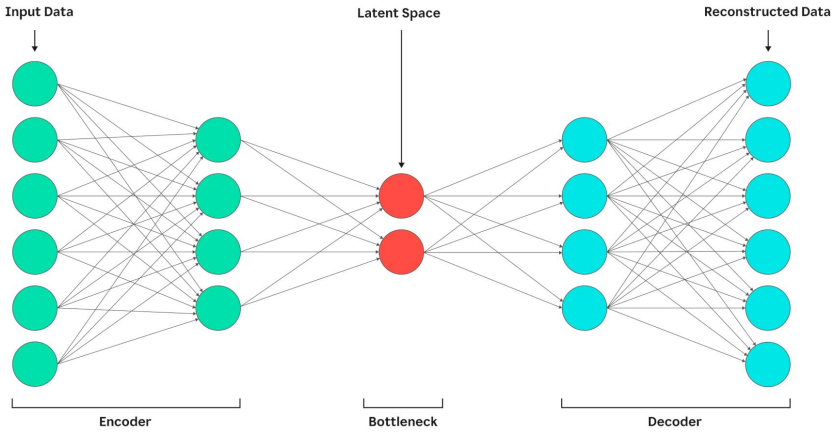
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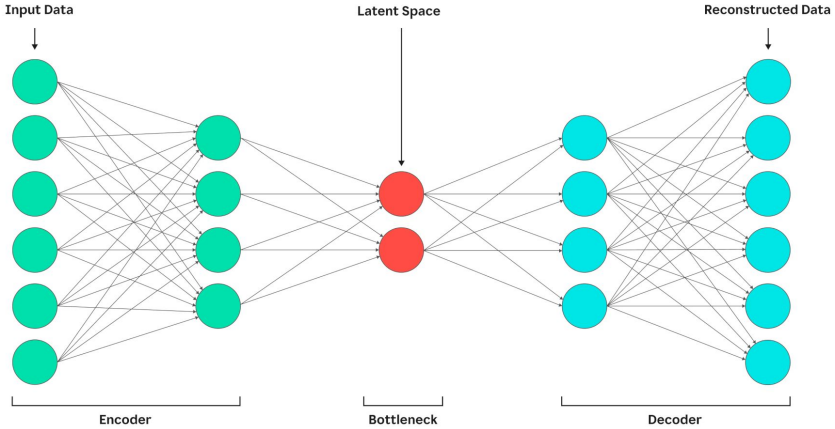
La función *f* es el encoder y la función *g* es el decoder.



Autoencoders

La función *f* es el encoder y la función *g* es el decoder.

Por lo tanto puede aprender una representación no lineal mejor que PCA.

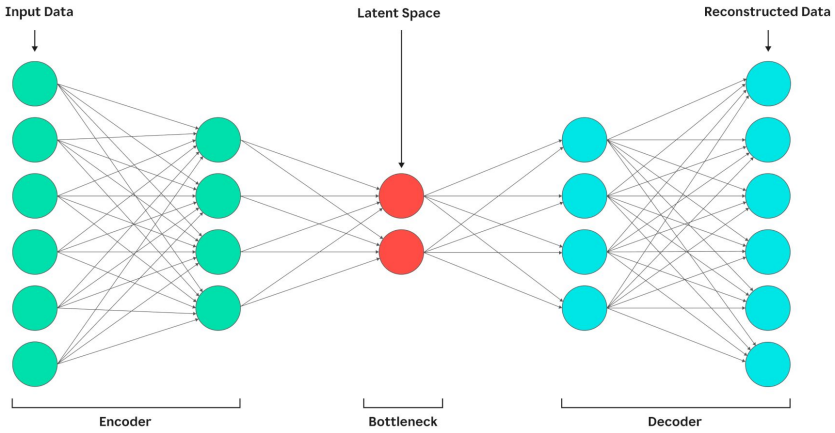


Autoencoders esparsos

Se agrega una regularización donde se penaliza si h es denso

$$f(x) = h$$

$$L(x, g(f(x))) + \Omega(h)$$



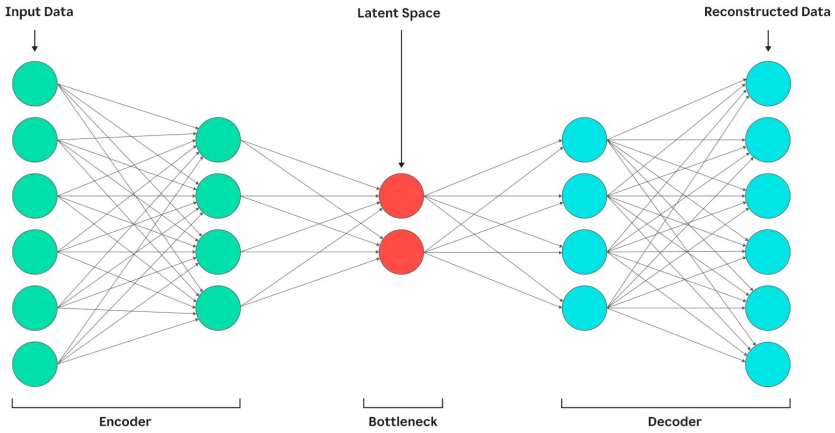
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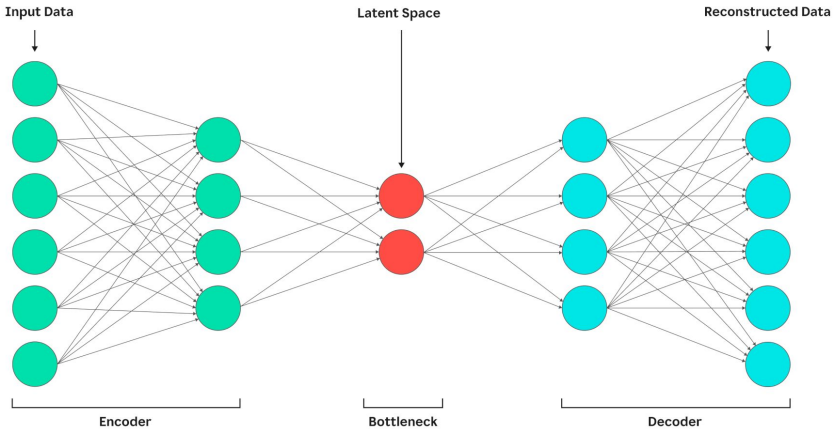
De esta forma, se aprovecha al máximo el tamaño de h



Denoising Autoencoders (DAE)

Autoencoders: Si la dimensión de h es igual a x, f y g deben ser la identidad ya que se minimiza:

$$L(x, g(f(x)))$$



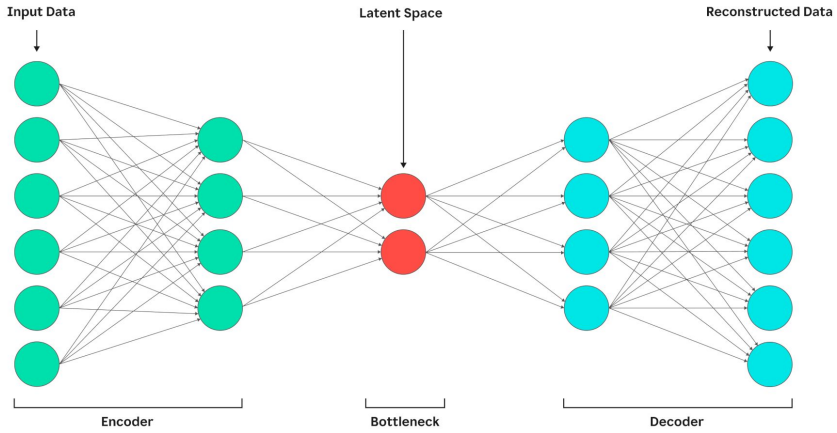
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Para evitar esto, Denoising autoencoders le agregan un ruido a x:

$$L(x, g(f(\hat{x})))$$



Denoising Autoencoders (DAE)

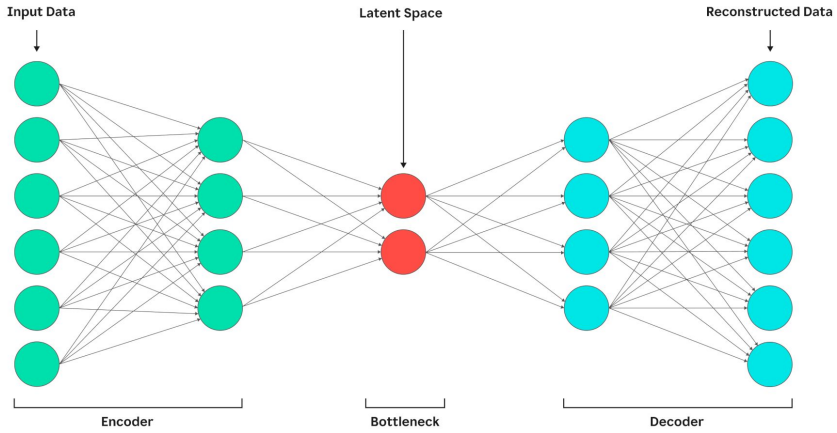
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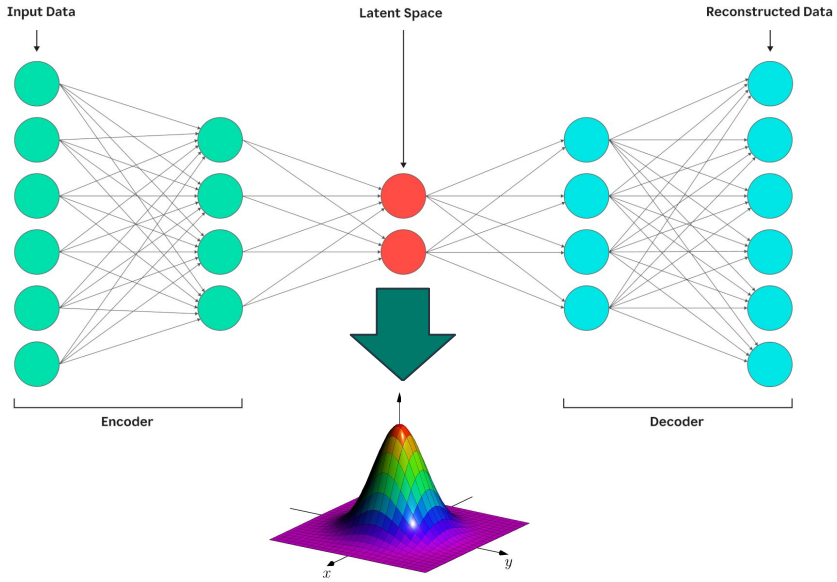
$$L(x, g(f(\hat{x})))$$

De esta forma, el autoencoder deber aprender a eliminar el ruido, mejorando la performance.



Variational Autoencoders (VAE)

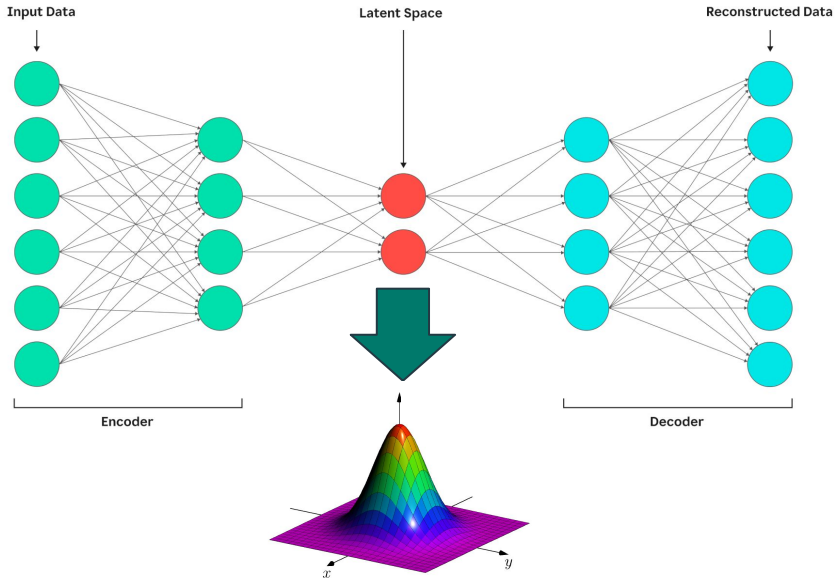
A diferencia de los Autoencoders, aprenden una representación continua.



Variational Autoencoders (VAE)

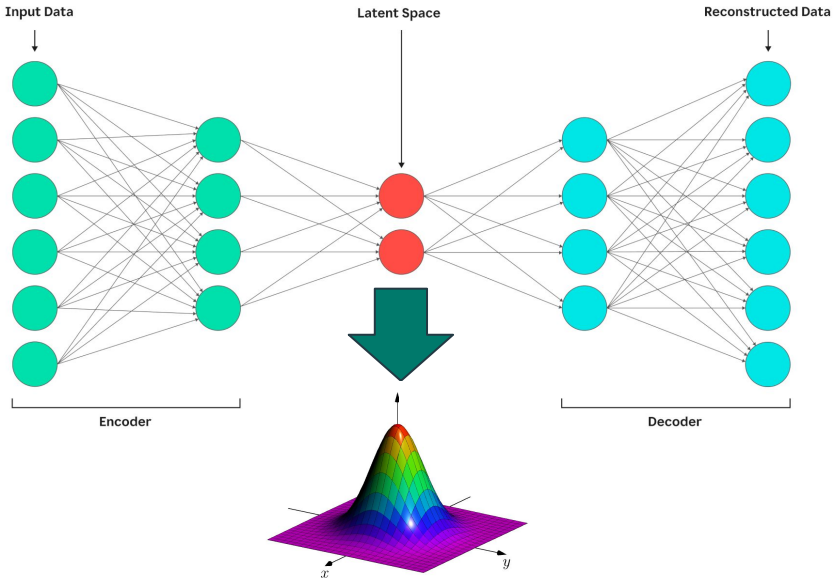
A diferencia de los Autoencoders, aprenden una representación continua.

No usan las variables latentes de los datos de entrenamiento como un valor discreto fijo z, sino como un rango continuo de posibilidades, expresado como una distribución de probabilidad p(z).



Variational Autoencoders (VAE)

Solo minimizando la loss, el modelo no organiza el espacio latente.

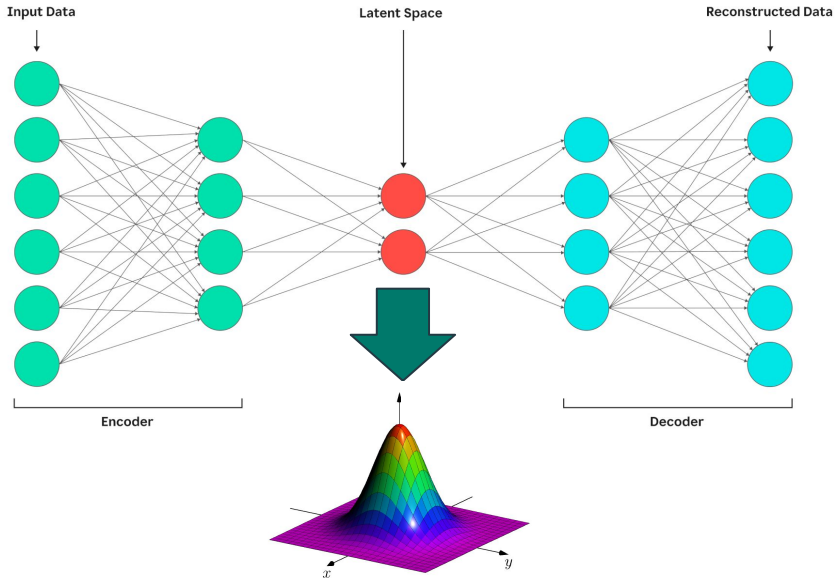


Variational Autoencoders (VAE)

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La **divergencia KL** es una métrica utilizada para comparar dos distribuciones de probabilidad.

Minimizar la divergencia KL entre la distribución aprendida de las variables latentes y una distribución gaussiana fuerza a que las variables latentes sigan una distribución normal.



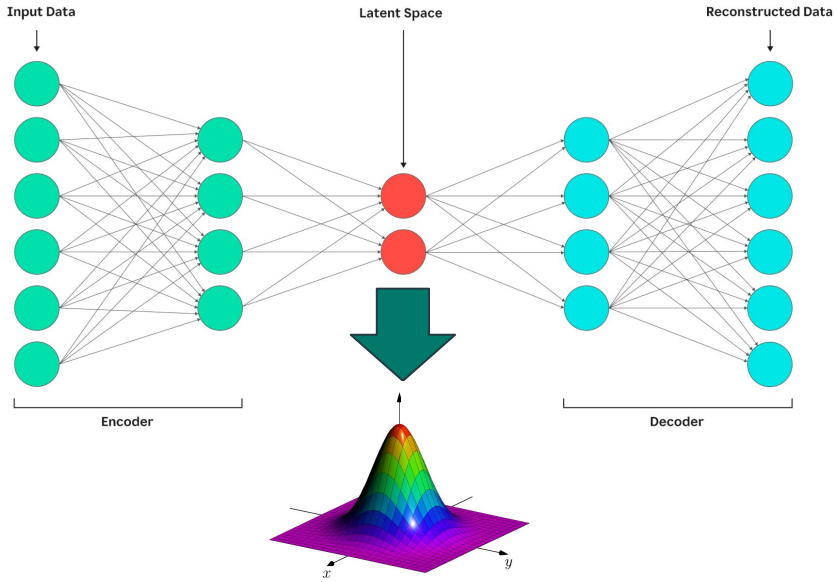
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Esto permite una interpolación suave de cualquier punto en el espacio latente.



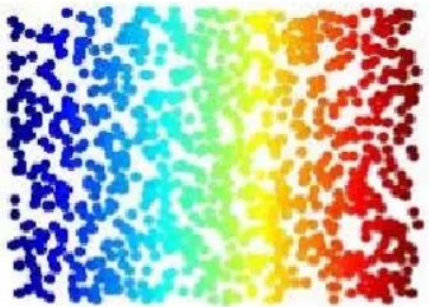
Manifolds y Autoencoders

Manifolds y Autoencoders

Manifold: Una estructura de baja dimensionalidad incrustada en un espacio de alta dimensionalidad.



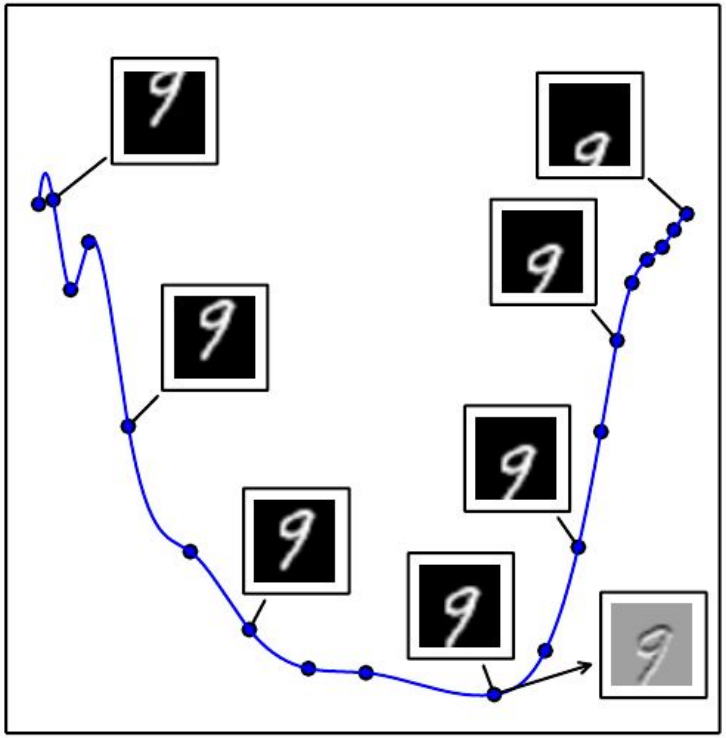
(a) Swiss roll



(b)Original manifold

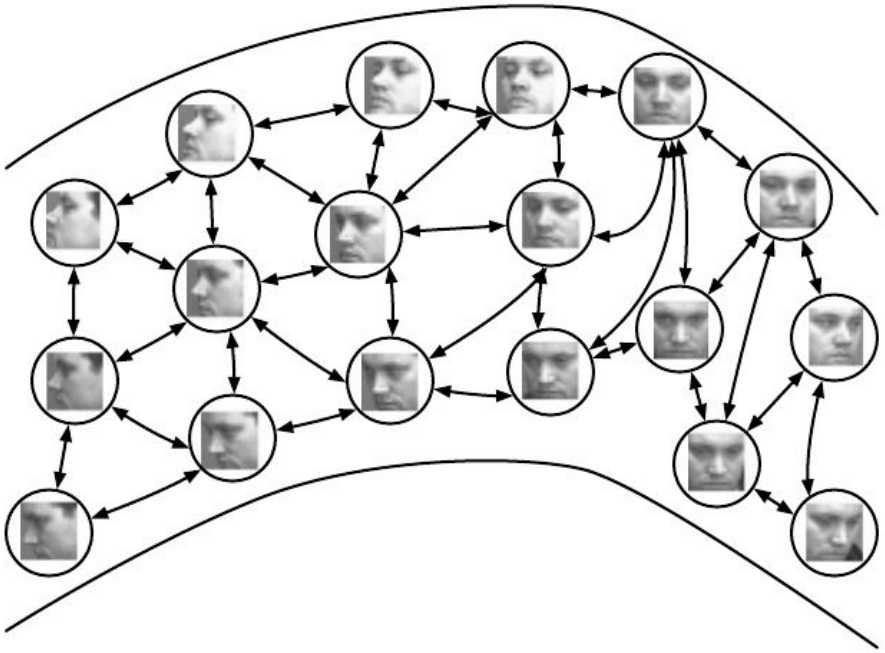
Manifolds y Autoencoders

Usando el dataset MINST (imagenes de numeros), uno puede encontrar una trayectoria en el espacio multidimensional que representa desplazamientos verticales.



Manifolds y Autoencoders

A partir de representaciones en menor dimensión de fotos, se puede ordenar según hacia donde mira la persona.



Qué vimos hoy

- Nociones preliminares de embeddings
- Word2Vec
- Autoencoders